Deep Podder

+91 6289622872 Kolkata, India conect.deeppodder@gmail.com linkedin.com/in/deeppodder2005
github.com/unknown-spec10

OBJECTIVE

Backend and AI engineer building production-grade RAG systems, async FastAPI backends, and cloud-integrated data pipelines. Seeking AI engineering or backend roles where I can ship systems that solve real problems.

EDUCATION

Bachelor of Computer Applications (BCA) — Institute of Engineering and Management, Kolkata — Graduating 2026

SKILLS

Backend	Python, FastAPI, Async Python (asyncpg, aioboto3), REST API design, WebSockets, SSE streaming, JWT & OAuth2, Celery, Redis, APScheduler
AI / ML	Large Language Models (LLMs), RAG pipelines, semantic caching, parent-child chunking, reranking (FlashRank), LangChain, LangGraph, LlamaIndex, hallucination mitigation, multimodal embeddings, prompt engineering
Databases	PostgreSQL, Vector Databases (pgvector), ChromaDB, Redis, MySQL, JSONB, indexing, query optimisation, ACID transactions, Alembic migrations
Cloud & Infra	AWS (EC2, S3 SSE-KMS, RDS, SNS, Cost Explorer), Docker, Supabase, Vercel, CI/CD-adjacent deployment pipelines
LLM APIs	Gemini 2.5 Flash, gemini-embedding-2-preview, Groq (LLaMA 3.3 70b), Anthropic API
Tools	Git, pdfplumber, rapidfuzz, Streamlit, watchdog, httpx, Fernet encryption, Agentic workflows

PROJECTS

Career Guidance AI (Career Sync)

FastAPI · pgvector · Groq · Gemini · React · Docker

- Architected a multi-stage resume parsing pipeline (pdfplumber + Gemini Vision fallback) with semaphore-controlled `asyncio.gather` firing 6 parallel section-extraction prompts, cutting wall time by **70% (12s → 3.5s)**.
- Refactored the job recommendation engine to eliminate N+1 queries, reducing SQL queries per run by **98.7% (301 → 4)**, and built a dual-layer embedding cache cutting Gemini API calls by **~99%**.
- Shipped a **51,000-line** backend with **91 REST endpoints** and **31 DB tables** (106-test suite, 97% pass rate); offloaded blocking LLM/SSE calls to background threads, holding **<10ms** latency on admin routes under concurrent load.

Insurance Policy Document Handler

FastAPI · AWS RDS/S3/SNS · Redis · PostgreSQL

- Architected a high-concurrency KYC document portal (FastAPI, asyncpg, aioboto3) using `asyncio.gather` for concurrent S3 presigned URL generation, cutting multi-document view latency by **99.7% (776ms → 2.3ms)**.
- Implemented Redis-based admin dashboard caching with SCAN-driven invalidation and composite indexes, dropping RDS query latency from **1,178ms to <1ms (1000x+ speedup)**.

- Designed SHA-256 hash-based duplicate upload blocking (saving up to 100% storage cost on duplicates) and automated 5-year retention cleanup across an 8,300-line, 27-endpoint system.

JiraSync — Jira-to-Power BI ETL Pipeline FastAPI · PostgreSQL · APScheduler · asyncpg

- Designed and deployed an async ETL service replacing a fragile Appfire OData connector, eliminating Power BI refresh timeouts and blank reports via a local PostgreSQL warehouse achieving **sub-3-second** incremental sync runs.
- Engineered API resilience with 10 req/s client-side pacing, 5-attempt exponential backoff, and PostgreSQL advisory locks to prevent race conditions between overlapping scheduler jobs.
- Built automated schema-drift monitoring with Slack/Teams alerts and JSONB-based flexible field storage (**sub-12ms** nested query latency), cutting Jira API payload size by **~90%**.